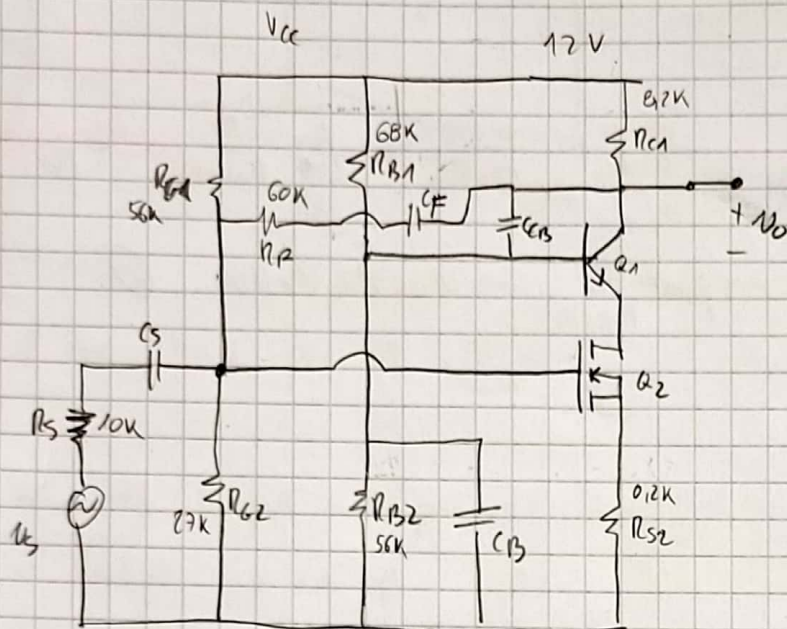
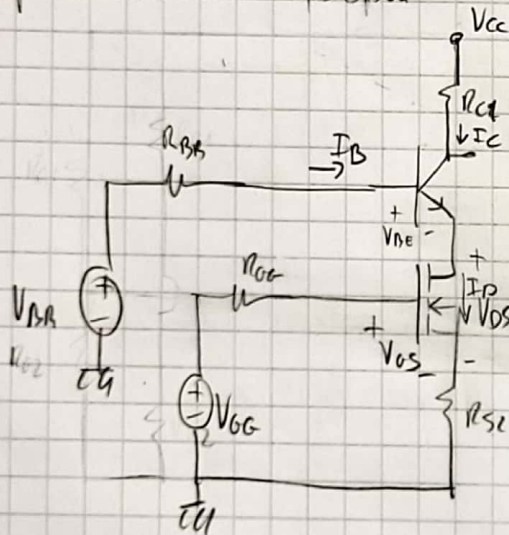




1) q analysis all caps open



$$V_{B1} = \frac{V_{CC} R_{B2}}{R_{B2} + R_{B1}} = 5.62V \quad R_{B1} = R_{B2} \parallel R_{B1} = 30.71k\Omega$$

$$V_{G1} = \frac{V_{CC} R_{G2}}{R_{G2} + R_{G1}} = 3.9V \quad R_{G1} = R_{G2} \parallel R_{G1} = 18.21k\Omega$$

$$\begin{cases} V_{G1} - V_{GS} - R_{S1} I_D = 0 \\ I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 \end{cases} \quad \begin{cases} V_{GS} = 3.75V \\ I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 = 0.765mA \end{cases}$$

$$I_{D1} = I_{D2} = (1 + \beta_{F1}) I_B \rightarrow I_B = 3.8\mu A$$

$$I_C = 0.761mA$$

$$V_{DS} = V_{B1} - R_{D1} I_C - V_{BE} - R_{S2} I_D = 4.45V$$

$$V_{GS} = V_{CC} - R_{D1} I_C - V_{DS} - R_{S2} I_D = 1.15V > 0.1V \text{ (error)}$$

$$g_{m1} = 20.27mS^{-1}$$

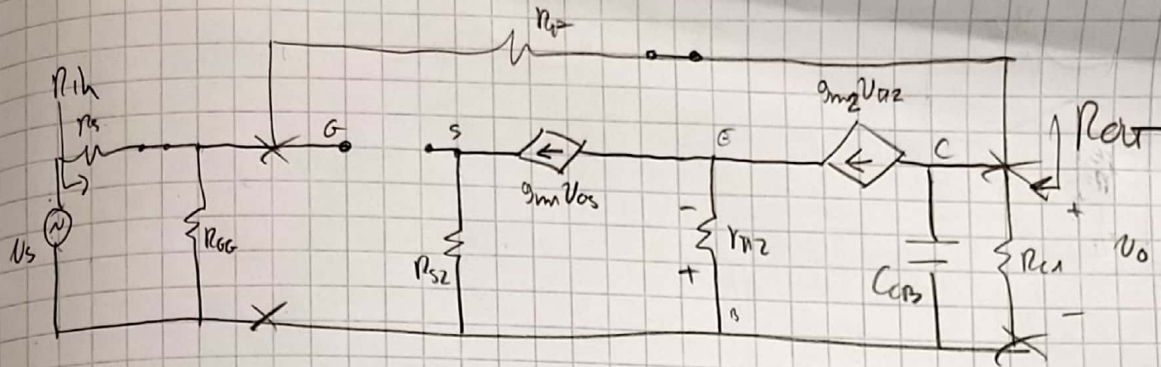
$$g_{m2} = 0.874mS^{-1}$$

$$r_{\pi} = 6.83k\Omega$$

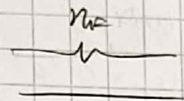
$$P_{Q1} = V_{B1} I_{B1} + V_{CC} I_C = 0.878mW$$

$$P_{Q2} = V_{DS} I_D = 3.4mW$$

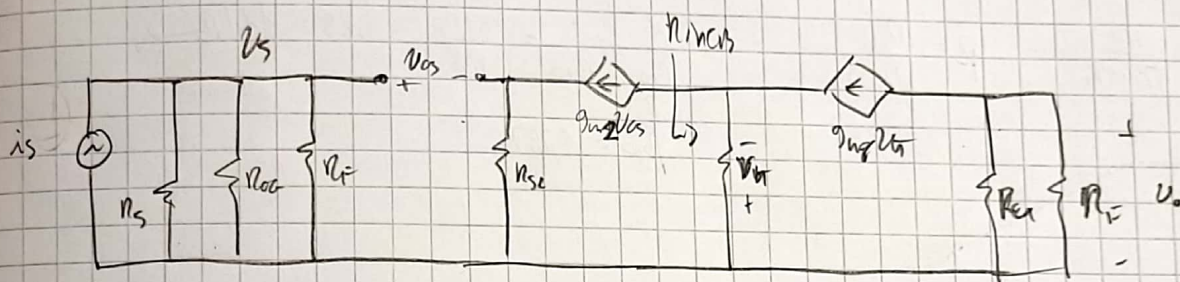
3) $A_v = v_o / v_s$ C_s, C_F, C_{m3} short C_{c1} open



current mirror voltage source



$$R_{UF} = R_{UF} = R_F \quad \beta = -1/\mu_{F}$$



$$R_{inB} = \frac{r_a}{1 + \beta_{02}} = 34 \Omega$$

$$A_{OL} = \frac{v_o}{v_s} = \frac{v_o}{v_A} \cdot \frac{v_A}{v_s} \cdot \frac{v_s}{i_s}$$

$$= g_{m1} (R_{D1} \parallel R_F) \cdot \frac{-g_{m2} R_{inB}}{1 + g_{m2} R_{S2}} \cdot R_S \parallel R_{G1} \parallel R_{G2} =$$

$$= 110,74 \cdot (-0,0253) \cdot 5,82 \text{ k}\Omega = -31,03 \text{ k}\Omega$$

$$A^F = \frac{A_{OL}}{1 + A_{OL}} = -20,65 \text{ k}\Omega$$

4) R_{in} :

$$R_{in}^{OL} = 5,82 \text{ k}\Omega$$

$$R_{in}^F = 3,83 \text{ k}\Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} + \frac{1}{R_s}} = 16,2 \text{ k}\Omega$$

$$5) R_{out} = R_F \parallel R_{D1} = 7,2 \text{ k}\Omega$$

$$R_{out}^F = 4,75 \text{ k}\Omega$$

6) due to C_{C1}

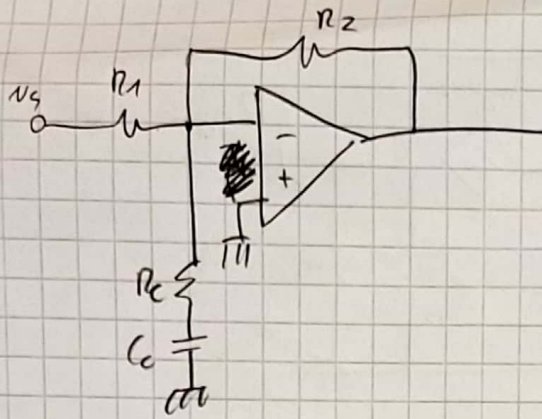
$$R_{C1}^0 = R_{C1} \parallel R_F = 7,2 \text{ k}\Omega$$

decreased by feedback

$f_H \uparrow$ on feedback

$$f_H = (1 + \beta_{OL}) \left(\frac{1}{C_{C1}} + \frac{1}{C_{C1} R_{C1}^0} \right) = 1,52 \text{ MHz}$$

CS 2)



1) inverting $A_v(s) = \frac{v_o}{v_s} = -\frac{R_2}{R_1} = -2,48$

$\omega_0 = 10^2$
 $\omega_1 = 2 \cdot 10^4$

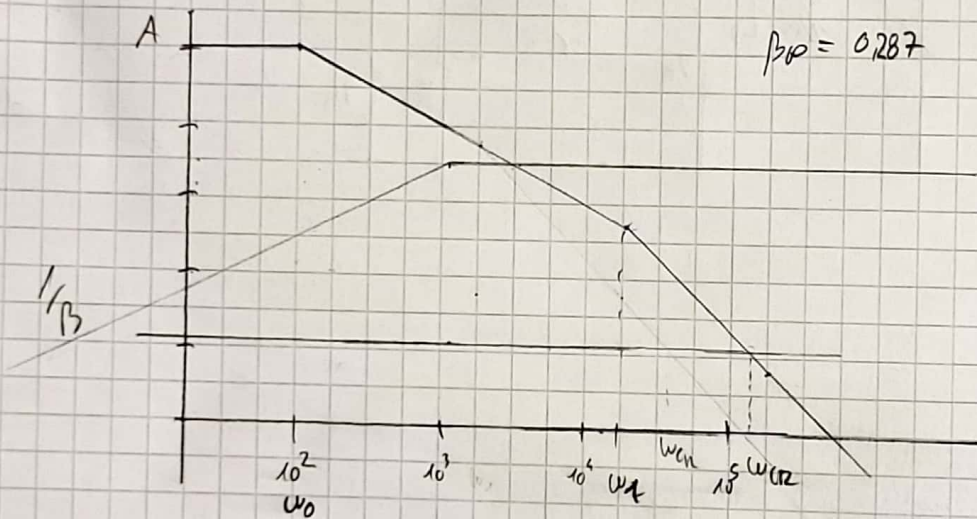
2)

$T = \frac{v_r}{v_e} = A(s) \cdot \frac{R_1}{R_1 + R_2}$

$\beta = \frac{R_1}{R_1 + R_2}$

$\frac{1}{\beta_0} = \frac{1}{\beta_{\infty}} = \frac{R_1 + R_2}{R_1} = 3,48 = 11 \text{ dB}$

$\beta_{10} = 0,287$



$A_{10} \omega_0 = A_{\infty} \omega_1$

$A_{\infty} \omega_1^2 = A_{10} \omega_0 \omega_c \beta_{10}$

$\omega_{cr} = \sqrt{A_{10} \omega_0 \omega_c \beta_{10}} = 2,4 \cdot 10^5 \text{ rad/s}$

$\beta_{10} \approx \omega_{cr} = 76,4 \text{ MHz}$

$\varphi_m = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_1}\right) = 4,79^\circ$

ident $A(s) =$

$A(s) = -\frac{R_2}{R_1} = -\frac{R_2}{R_1 // \frac{1+sC R_c}{sC}}$

3) here φ_m

zero is introduced at $\omega_z = \frac{1}{C R_c} = 1,377 \cdot 10^3 \text{ rad/s}$

and $\omega_p = \frac{1}{C(R_c + R_1 + R_2)} = 193 \text{ rad/s}$

corresponding $\beta_{10} = \frac{R_1 R_c}{R_2 + R_1 R_c} \approx \frac{R_c}{R_2 + R_1} \rightarrow \frac{1}{\beta_{10}} = 68 \text{ dB}$

$A_{10} \omega_0 = \frac{1}{\beta_{10}} \omega_{cr} \rightarrow \omega_{cr} = A_{10} \omega_0 \beta_{10} = 6022 \text{ rad/s}$

$\varphi_m = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_1}\right) + \arctan\left(\frac{\omega_{cr}}{\omega_z}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_p}\right) = 64,7^\circ$

Very narrow BW!